

Text Recognition

Preprocessing Techniques:

- Isolate text from complex backgrounds
- Denoising
- Deskewing
- CLAHE (Contrast Enhancement)

This is done to enhance the text detection accuracy and improved readability for the computer vision models

Poster Preparation

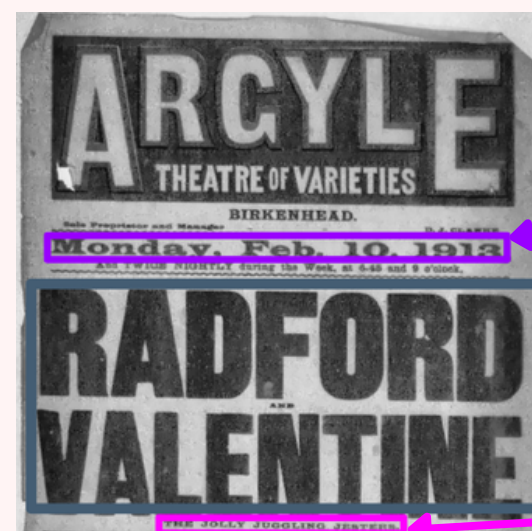


LLMs Used



Text Extraction

LLMs are prompted to read and transcribe the victorian posters into structured metadata and format (in CSV and JSON) while preserving their original artistic phrasing

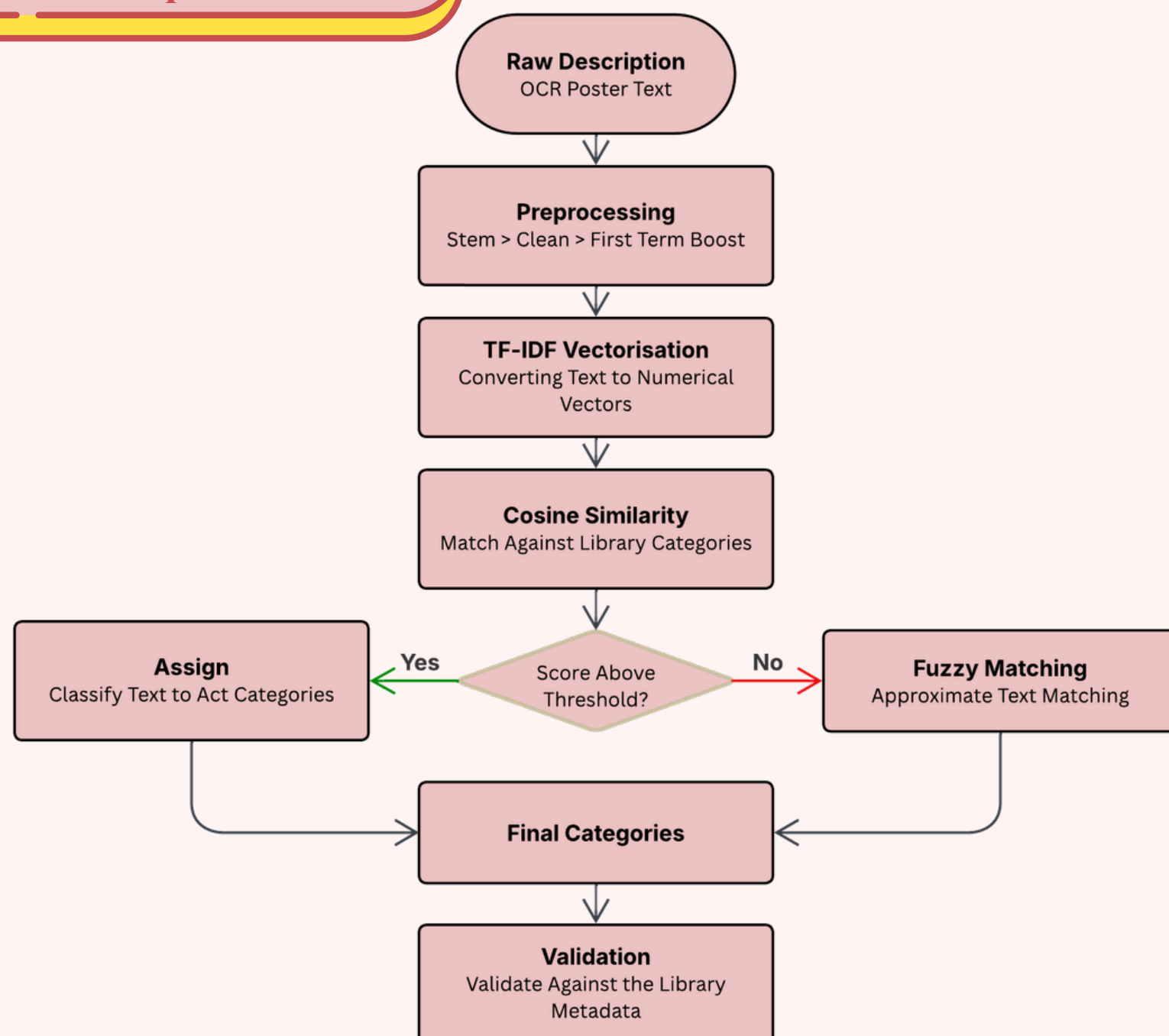


Date

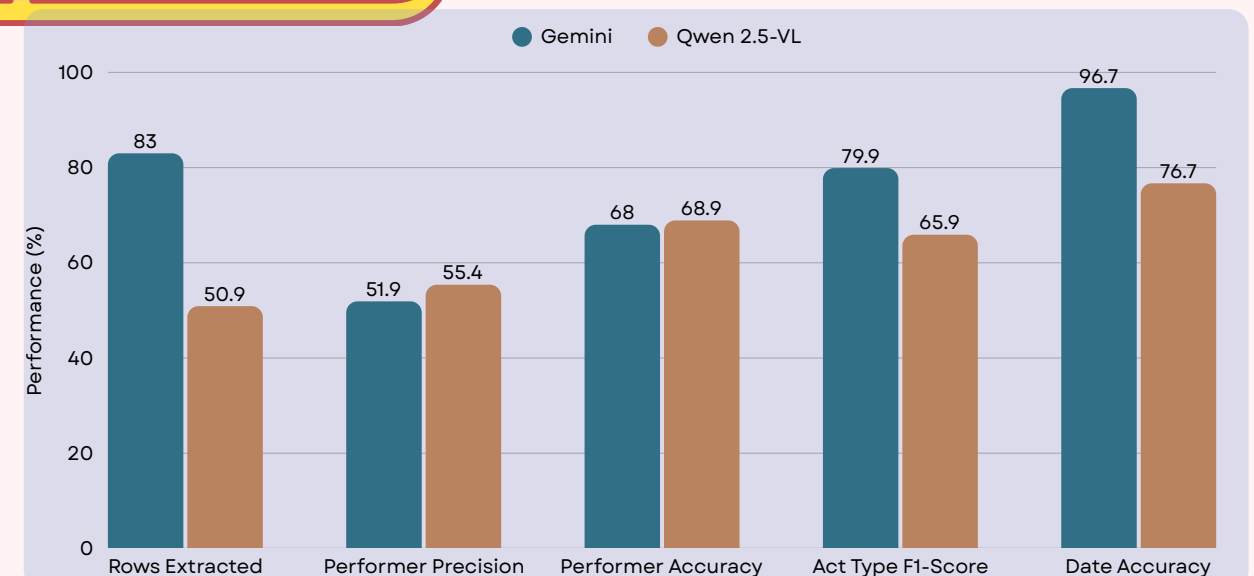
Performers

Act Name

Act Type Classification Pipeline



LLM Comparisons



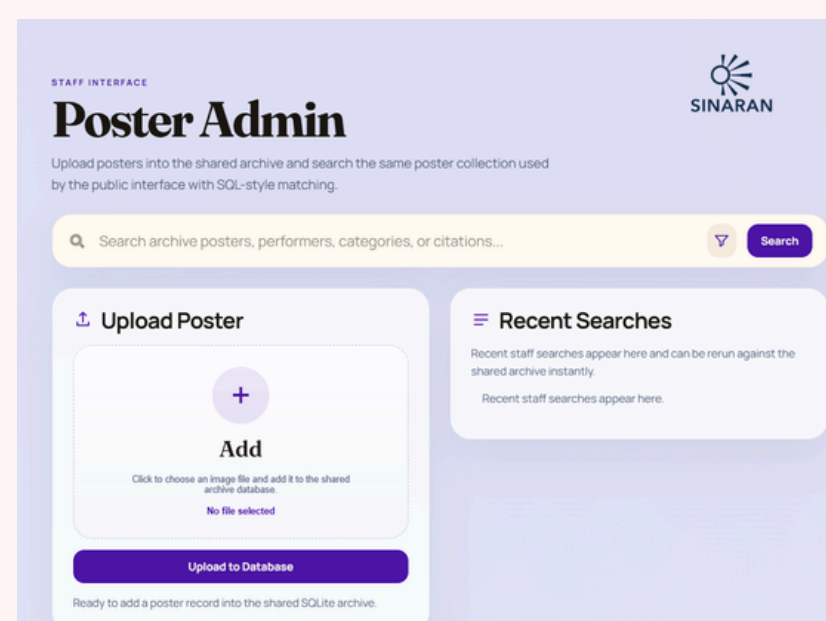
Gemini consistently outperformed Qwen in all metrics. Gemini is preferred for high-accuracy historical data extraction when cloud resources are available, though Qwen remains a viable resource-efficient alternative.

Metric (Act Type)	Gemini 2.5 Pro	Qwen 2.5-VL
Precision	73.80%	59.80%
Recall	95.80%	76.20%
Weighted F1-Score	79.90%	65.90%

Upload To Software

The prototype supports batch uploads of the existing poster data, while authorised admin users can also add new entries into the database.

To optimise processing, the software queues uploads and triggers the LLMs once a buffer of 10 posters is reached



Filtered Searches

The software features predictive search and personalised recommendations, including trending and recent queries.

By utilising semantic linking, the system identifies related terms to suggest the posters most relevant to the user's search.

